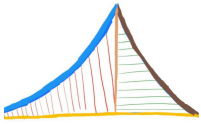
# Math for EE



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## About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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# 1. CBSE 2019

## A. Formula-10

1.1.1 Distance Formula and Equidistant Locus. The Euclidean distance between two points $\mathbf{x}_1 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ and $\mathbf{x}_2 = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ in $\mathbb{R}^2$ is given by

$$\|\mathbf{x}_1 - \mathbf{x}_2\|^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2 \tag{1.1.1.1}$$

A point $\mathbf{x}$ is equidistant from two fixed points $\mathbf{a}$ and $\mathbf{b}$ if

$$\|\mathbf{x} - \mathbf{a}\|^2 = \|\mathbf{x} - \mathbf{b}\|^2 \tag{1.1.1.2}$$

**Solution:** Expanding (1.1.1.2):

$$\|\mathbf{x}\|^2 - 2\mathbf{a}^\top \mathbf{x} + \|\mathbf{a}\|^2 = \|\mathbf{x}\|^2 - 2\mathbf{b}^\top \mathbf{x} + \|\mathbf{b}\|^2 \tag{1.1.1.3}$$

$$\implies 2(\mathbf{b} - \mathbf{a})^\top \mathbf{x} = \|\mathbf{b}\|^2 - \|\mathbf{a}\|^2 \tag{1.1.1.4}$$

yielding the equation of the perpendicular bisector.

1.1.2 Standard Trigonometric Values and Complementary Identity:

$$\sin(60^\circ) = \frac{\sqrt{3}}{2}, \quad \tan(45^\circ) = 1, \quad \cos(30^\circ) = \frac{\sqrt{3}}{2} \tag{1.1.2.1}$$

For complementary angles:

$$\cos(90^\circ - \theta) = \sin\theta \implies \cos(30^\circ) = \sin(60^\circ) \tag{1.1.2.2}$$

1.1.3 Fundamental Pythagorean Trigonometric Identities:

$$\cos^2 A + \sin^2 A = 1 \tag{1.1.3.1}$$

**Solution:** From (1.1.3.1), solving for $\cos A$ and its reciprocal $\sec A$:

$$\cos A = \sqrt{1 - \sin^2 A} \tag{1.1.3.2}$$

$$\sec A = \frac{1}{\cos A} = \frac{1}{\sqrt{1 - \sin^2 A}} \tag{1.1.3.3}$$

1.1.4 Pythagoras Theorem. In a right-angled triangle with legs $a, b$ and hypotenuse $c$:

$$c^2 = a^2 + b^2 \tag{1.1.4.1}$$

For an isosceles right triangle where $a = b$:

$$c = \sqrt{a^2 + a^2} = a\sqrt{2} \tag{1.1.4.2}$$

1.1.5 Basic Proportionality Theorem (Thales' Theorem). If a line is drawn parallel to one side of a triangle intersecting the other two sides, then it divides the two sides in the same ratio:

$$\frac{AD}{BD} = \frac{AE}{CE} \tag{1.1.5.1}$$

**Solution:** Let $h_1$ be the altitude from $E$ to $AB$ and $h_2$ be the altitude from $D$ to

$AC$.

$$\frac{\text{ar}\,(\triangle ADE)}{\text{ar}\,(\triangle BDE)} = \frac{\frac{1}{2}AD \cdot h_1}{\frac{1}{2}BD \cdot h_1} = \frac{AD}{BD} \tag{1.1.5.2}$$

$$\frac{\text{ar}\,(\triangle ADE)}{\text{ar}\,(\triangle CDE)} = \frac{\frac{1}{2}AE \cdot h_2}{\frac{1}{2}CE \cdot h_2} = \frac{AE}{CE} \tag{1.1.5.3}$$

Since $\triangle BDE$ and $\triangle CDE$ share base $DE$ between parallel lines $DE \parallel BC$, $\text{ar}\,(\triangle BDE) = \text{ar}\,(\triangle CDE)$, yielding (1.1.5.1).

1.1.6 Discriminant and Nature of Quadratic Roots. For the general quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$), the discriminant $D$ is defined as

$$D = b^2 - 4ac \tag{1.1.6.1}$$

- $D > 0$: Two distinct real roots.
- $D = 0$: Two equal real roots.
- $D < 0$: No real roots (complex conjugate pair).

1.1.7 Product and GCD/LCM Duality for Two Positive Integers:

$$\gcd\,(a, b) \times \text{lcm}\,(a, b) = a \times b \tag{1.1.7.1}$$

**Solution:** Let $a = \prod p_i^{\alpha_i}$ and $b = \prod p_i^{\beta_i}$. Then:

$$\gcd\,(a, b) = \prod p_i^{\min(\alpha_i, \beta_i)}, \quad \text{lcm}\,(a, b) = \prod p_i^{\max(\alpha_i, \beta_i)} \tag{1.1.7.2}$$

Since $\min\,(\alpha_i, \beta_i) + \max\,(\alpha_i, \beta_i) = \alpha_i + \beta_i$, multiplying yields (1.1.7.1).

1.1.8 Sum of the First $n$ Terms of an Arithmetic Progression:

$$S_n = \frac{n}{2}\left[2a + (n-1)\,d\right] \tag{1.1.8.1}$$

**Solution:** Writing $S_n$ forward and backward:

$$S_n = a + (a+d) + \cdots + (a + (n-1)d) \tag{1.1.8.2}$$

$$S_n = (a + (n-1)d) + \cdots + (a+d) + a \tag{1.1.8.3}$$

Summing the two equations term-by-term:

$$2S_n = n\left[2a + (n-1)\,d\right] \tag{1.1.8.4}$$

yielding (1.1.8.1).

1.1.9 Consistency and Infinite Solutions of Linear Systems. For the matrix system $\mathbf{Ax} = \mathbf{b}$, infinite solutions exist if and only if

$$\text{rank}\,(\mathbf{A}) = \text{rank}\,([\mathbf{A} \mid \mathbf{b}]) < n \tag{1.1.9.1}$$

where $n$ is the number of unknown variables.

1.1.10 Mode of Grouped Frequency Distribution:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h \tag{1.1.10.1}$$

where:

- $l$ is the lower limit of the modal class
- $h$ is the class interval width
- $f_1$ is the frequency of the modal class
- $f_0$ is the frequency of the preceding class
- $f_2$ is the frequency of the succeeding class

1.1.11 Classical Probability and Partition Axiom. If a sample space $S$ is partitioned into pairwise disjoint events $E_1, E_2, \ldots, E_k$:

$$\sum_{i=1}^{k} \Pr(E_i) = 1, \quad \text{where } \Pr(E) = \frac{n(E)}{n(S)} \tag{1.1.11.1}$$

1.1.12 Collinearity of Three Points in $\mathbb{R}^2$. Points $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are collinear if and only if their displacement vectors are linearly dependent:

$$\begin{vmatrix} \mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A} \end{vmatrix} = 0 \tag{1.1.12.1}$$

1.1.13 Area of a Triangle using Vector Determinant:

$$\text{Area}(\triangle ABC) = \frac{1}{2} \left| \begin{vmatrix} \mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A} \end{vmatrix} \right| \tag{1.1.13.1}$$

1.1.14 Section Formula in Vector Notation. The point $\mathbf{P}$ dividing the segment between $\mathbf{A}$ and $\mathbf{B}$ internally in ratio $m : n$ is given by

$$\mathbf{P} = \frac{m\mathbf{B} + n\mathbf{A}}{m+n} \tag{1.1.14.1}$$

1.1.15 Matrix Inverse and Cramer's Rule for $2 \times 2$ Systems. If $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $\left|\mathbf{M}\right| = ad - bc \neq 0$:

$$\mathbf{M}^{-1} = \frac{1}{\left|\mathbf{M}\right|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \tag{1.1.15.1}$$

The unique solution to $\mathbf{M}\mathbf{x} = \mathbf{b}$ is $\mathbf{x} = \mathbf{M}^{-1}\mathbf{b}$.

1.1.16 Euclidean Algorithm via Elementary Shear Transformations. The reduction of an integer vector $\mathbf{v}_0 = \begin{pmatrix} a \\ b \end{pmatrix}$ using shear matrices $\mathbf{E}_k = \begin{pmatrix} 1 & -q_k \\ 0 & 1 \end{pmatrix}$ or $\begin{pmatrix} 1 & 0 \\ -q_k & 1 \end{pmatrix}$ satisfies

$$\left|\mathbf{E}_k\right| = 1 \implies \gcd(a,b) = \gcd(\mathbf{E}_k \mathbf{v}_0) \tag{1.1.16.1}$$

terminating when one coordinate becomes zero, leaving the non-zero invariant as $\gcd(a,b)$.

1.1.17 Intersecting Chords Theorem. If two chords $AC$ and $BD$ of a circle intersect at an interior point $P$:

$$AP \times PC = BP \times DP \tag{1.1.17.1}$$

**Solution:** In $\triangle PAB$ and $\triangle PDC$, angles in the same circular segment are equal

($\angle PBA = \angle PCD$) and vertically opposite angles are equal ($\angle APB = \angle DPC$). By AA similarity:

$$\triangle PAB \sim \triangle PDC \implies \frac{AP}{DP} = \frac{BP}{PC} \tag{1.1.17.2}$$

Cross-multiplying yields (1.1.17.1).

1.1.18 Ratio of Areas of Similar Triangles. If $\triangle ABC \sim \triangle DEF$, then

$$\frac{\text{ar}\,(\triangle ABC)}{\text{ar}\,(\triangle DEF)} = \left(\frac{AB}{DE}\right)^2 = \left(\frac{BC}{EF}\right)^2 = \left(\frac{CA}{FD}\right)^2 \tag{1.1.18.1}$$

1.1.19 Tangent Hyperplane to a Circle. The tangent line to the circle $\|\mathbf{x}\| = r$ at contact point $\mathbf{r}_C = \begin{pmatrix} r\cos\theta \\ r\sin\theta \end{pmatrix}$ has normal vector $\hat{n} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$ and equation

$$\hat{n}^\top \mathbf{x} = r \iff x\cos\theta + y\sin\theta = r \tag{1.1.19.1}$$

Two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if

$$\mathbf{u}^\top \mathbf{v} = 0 \iff \mathbf{u} \perp \mathbf{v} \tag{1.1.19.2}$$

1.1.20 Mensuration of Standard Plane and Composite Figures:

$$\text{Area of quadrant} = \frac{1}{4}\pi r^2 \tag{1.1.20.1}$$

$$\text{Area of square} = s^2, \quad \text{Diagonal } d = s\sqrt{2} \tag{1.1.20.2}$$

$$\text{Area of circumscribed circle} = \pi R^2 = \pi\left(\frac{s\sqrt{2}}{2}\right)^2 = \frac{\pi s^2}{2} \tag{1.1.20.3}$$

1.1.21 Mensuration of Cylindrical and Spherical Solids:

$$V_{\text{cylinder}} = \pi r^2 h \tag{1.1.21.1}$$

$$V_{\text{hemisphere}} = \frac{2}{3}\pi r^3 \implies V_{\text{sphere}} = \frac{4}{3}\pi r^3 \tag{1.1.21.2}$$

$$\text{Mass} = \rho \times V \tag{1.1.21.3}$$

where $\rho$ is the mass density and $V$ is the total volume.

1.1.22 Arithmetic Mean of Grouped Data (Direct Method):

$$\bar{x} = \frac{\sum_{i=1}^{k} f_i x_i}{\sum_{i=1}^{k} f_i}, \quad \text{where } x_i = \frac{l_i + u_i}{2} \tag{1.1.22.1}$$

1.1.23 Polynomial Division and Quotient Ring Reduction. In the polynomial ring $\mathbb{R}[x]$ modulo ideal $\langle d(x)\rangle$:

$$f(x) \equiv r(x) \pmod{d(x)}, \quad \deg(r) < \deg(d) \tag{1.1.23.1}$$

If $d(x)$ divides $f(x)$, then $r(x) \equiv 0$.

1.1.24 Vieta's Relations for a Quadratic Polynomial. If $\alpha$ and $\beta$ are zeroes of $p(x) =$

$ax^2+bx+c$:

$$\alpha+\beta=-\frac{b}{a} \tag{1.1.24.1}$$

$$\alpha\beta=\frac{c}{a} \tag{1.1.24.2}$$

1.1.25 Algebraic Identities for Trigonometric Proofs:

$$a^3-b^3=(a-b)\left(a^2+ab+b^2\right) \tag{1.1.25.1}$$

$$\csc^2\theta-\cot^2\theta=1 \tag{1.1.25.2}$$

$$\sec^2\theta-\tan^2\theta=1 \tag{1.1.25.3}$$

*B. Class-10*

1.2.1 Write the coordinates of the point $P$ on the $x$-axis which is equidistant from the points $A\left(-2,0\right)$ and $B\left(6,0\right)$. (10 2019)

**Solution:** Any point on the $x$-axis has its $y$-coordinate equal to 0. Let the position vectors of the points be

$$\mathbf{P}=\begin{pmatrix}x\\0\end{pmatrix},\quad \mathbf{A}=\begin{pmatrix}-2\\0\end{pmatrix},\quad \mathbf{B}=\begin{pmatrix}6\\0\end{pmatrix} \tag{1.2.1.1}$$

Since $\mathbf{P}$ is equidistant from $\mathbf{A}$ and $\mathbf{B}$, from (1.1.1.1):

$$\|\mathbf{P}-\mathbf{A}\|^2=\|\mathbf{P}-\mathbf{B}\|^2 \tag{1.2.1.2}$$

From (1.1.1.2), evaluating the squared Euclidean distances:

$$\left(x-(-2)\right)^2+\left(0-0\right)^2=\left(x-6\right)^2+\left(0-0\right)^2 \tag{1.2.1.3}$$

$$\left(x+2\right)^2=\left(x-6\right)^2 \tag{1.2.1.4}$$

$$x^2+4x+4=x^2-12x+36 \tag{1.2.1.5}$$

$$16x=32\implies x=2 \tag{1.2.1.6}$$

Therefore, the coordinates of point $P$ are

$$\mathbf{P}=\begin{pmatrix}2\\0\end{pmatrix} \tag{1.2.1.7}$$

The following code verifies the solution in Fig. 1.2.1.1:

codes/10/q1.py

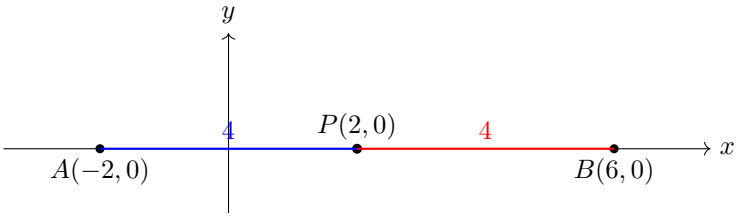



Fig. 1.2.1.1: Point $P(2,0)$ equidistant from $A(-2,0)$ and $B(6,0)$

1.2.2 Evaluate:

$$\sin^2(60°) + 2\tan(45°) - \cos^2(30°) \tag{1.2.2.1}$$

(10 2019)

**Solution:** From (1.1.2.1) and (1.1.2.2):

$$\sin(60°) = \frac{\sqrt{3}}{2}, \quad \tan(45°) = 1, \quad \cos(30°) = \frac{\sqrt{3}}{2} \tag{1.2.2.2}$$

Substituting these values into the expression:

$$\sin^2(60°) + 2\tan(45°) - \cos^2(30°) = \left(\frac{\sqrt{3}}{2}\right)^2 + 2(1) - \left(\frac{\sqrt{3}}{2}\right)^2 \tag{1.2.2.3}$$

$$= \frac{3}{4} + 2 - \frac{3}{4} = 2 \tag{1.2.2.4}$$

1.2.3 If $\sin A = \frac{3}{4}$, calculate $\sec A$. (10 2019)

**Solution:** From (1.1.3.2):

$$\cos A = \sqrt{1 - \sin^2 A} = \sqrt{1 - \left(\frac{3}{4}\right)^2} = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4} \tag{1.2.3.1}$$

From (1.1.3.3):

$$\sec A = \frac{1}{\cos A} = \frac{4}{\sqrt{7}} \tag{1.2.3.2}$$

1.2.4 In Fig. 1.2.4.1, $ABC$ is an isosceles triangle right-angled at $C$ with $AC = 4$ cm. Find the length of $AB$. (10 2019)

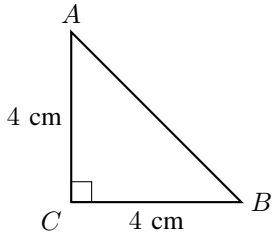



Fig. 1.2.4.1: Isosceles right triangle $ABC$

**Solution:** Triangle $ABC$ is an isosceles right-angled triangle at $C$. Since the hypotenuse $AB$ is the longest side, the two equal legs are $AC$ and $BC$:

$$BC = AC = 4 \text{ cm} \tag{1.2.4.1}$$

Applying the Pythagoras theorem (1.1.4.1) and (1.1.4.2):

$$AB^2 = AC^2 + BC^2 = 4^2 + 4^2 = 32 \tag{1.2.4.2}$$

$$\implies AB = \sqrt{32} = 4\sqrt{2} \text{ cm} \tag{1.2.4.3}$$

1.2.5 In Fig. 1.2.5.1, $DE \parallel BC$. Find the length of side $AD$, given that $AE = 1.8$ cm, $BD = 7.2$ cm, and $CE = 5.4$ cm. (10 2019)

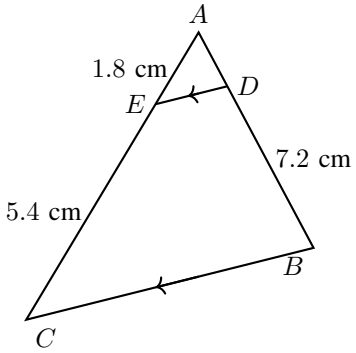



Fig. 1.2.5.1: Triangle $ABC$ with $DE \parallel BC$

**Solution:** In $\triangle ABC$, since $DE \parallel BC$, applying Thales' Theorem (1.1.5.1):

$$\frac{AD}{BD} = \frac{AE}{CE} \tag{1.2.5.1}$$

Substituting the given lengths:

$$\frac{AD}{7.2} = \frac{1.8}{5.4} = \frac{1}{3} \implies AD = \frac{7.2}{3} = 2.4 \text{ cm} \tag{1.2.5.2}$$

1.2.6 Find the nature of roots of the quadratic equation $2x^2 - 4x + 3 = 0$. (10 2019)

**Solution:** Comparing with the standard quadratic form $ax^2 + bx + c = 0$:

$$a = 2, \quad b = -4, \quad c = 3 \tag{1.2.6.1}$$

From (1.1.6.1), the discriminant $D$ is:

$$D = b^2 - 4ac = (-4)^2 - 4\,(2)\,(3) = 16 - 24 = -8 \tag{1.2.6.2}$$

Since $D < 0$, the quadratic equation has no real roots. The following code verifies the solution in Fig. 1.2.6.1:

```
codes/10/q6.py
```

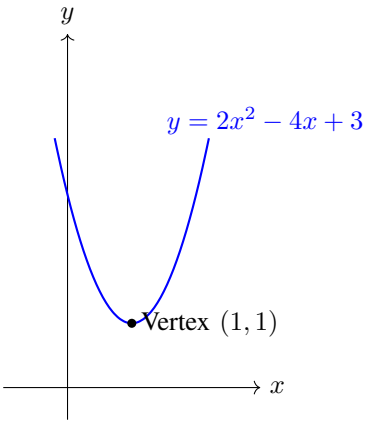



Fig. 1.2.6.1: Plot of $y = 2x^2 - 4x + 3$ showing no real roots ($D < 0$)

1.2.7 If $\gcd(336, 54) = 6$, find $\operatorname{lcm}(336, 54)$. (10 2019)

**Solution:** Let $a = 336$ and $b = 54$. From (1.1.7.1):

$$\gcd(a, b) \times \operatorname{lcm}(a, b) = a \times b \tag{1.2.7.1}$$

Rearranging:

$$\operatorname{lcm}(336, 54) = \frac{a \times b}{\gcd(a, b)} \tag{1.2.7.2}$$

$$= \frac{336 \times 54}{6} = 336 \times 9 = 3024 \tag{1.2.7.3}$$

1.2.8 Find the $21^{\text{st}}$ term of the A.P. $-4\frac{1}{2}, -3, -1\frac{1}{2}, \ldots$ (10 2019)

**Solution:** The first term is $a = -\frac{9}{2}$. The common difference $d$ is:

$$d = -3 - \left(-\frac{9}{2}\right) = -3 + \frac{9}{2} = \frac{3}{2} \tag{1.2.8.1}$$

From (**??**), the $n$-th term is:

$$a_n = a + (n - 1)\, d \tag{1.2.8.2}$$

Substituting $n = 21$:

$$a_{21} = -\frac{9}{2} + (21 - 1)\left(\frac{3}{2}\right) \tag{1.2.8.3}$$

$$= -\frac{9}{2} + 20\left(\frac{3}{2}\right) = -\frac{9}{2} + 30 = \frac{51}{2} = 25\frac{1}{2} \tag{1.2.8.4}$$

1.2.9 For what value of $k$ will the system of equations $2x + 3y = 7$ and $(k+2)x - 3(1-k)y = 5k + 1$ have infinitely many solutions? (10 2019)

**Solution:** Writing the system of linear equations in canonical matrix form $\mathbf{Ax} = \mathbf{b}$:

$$\begin{pmatrix} 2 & 3 \\ k+2 & 3k-3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 \\ 5k+1 \end{pmatrix} \tag{1.2.9.1}$$

The augmented matrix is

$$[\mathbf{A} \mid \mathbf{b}] = \begin{pmatrix} 2 & 3 & 7 \\ k+2 & 3k-3 & 5k+1 \end{pmatrix} \tag{1.2.9.2}$$

Performing the elementary row operation $R_2 \to 2R_2 - (k+2)R_1$:

$$[\mathbf{A} \mid \mathbf{b}] \xrightarrow{R_2 \to 2R_2-(k+2)R_1} \begin{pmatrix} 2 & 3 & 7 \\ 0 & 3k-12 & 3k-12 \end{pmatrix} \tag{1.2.9.3}$$

From (1.1.9.1), for the system to have infinitely many solutions:

$$\text{rank}\,(\mathbf{A}) = \text{rank}\,([\mathbf{A} \mid \mathbf{b}]) < 2 \tag{1.2.9.4}$$

This requires the second row to be entirely zero:

$$3k - 12 = 0 \implies k = 4 \tag{1.2.9.5}$$

1.2.10 Find the mode of the following frequency distribution:

| Class Interval | 25–30 | 30–35 | 35–40 | 40–45 | 45–50 | 50–55 |
|---|---|---|---|---|---|---|
| Frequency | 25 | 34 | 50 | 42 | 38 | 14 |

(10 2019)

**Solution:** The maximum frequency is $50$, corresponding to the modal class $35$–$40$. From the modal class:

$$l = 35, \quad h = 5, \quad f_1 = 50, \quad f_0 = 34, \quad f_2 = 42 \tag{1.2.10.1}$$

From (1.1.10.1), the mode is given by:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h \tag{1.2.10.2}$$

$$= 35 + \left(\frac{50 - 34}{2\,(50) - 34 - 42}\right) \times 5 \tag{1.2.10.3}$$

$$= 35 + \left(\frac{16}{24}\right) \times 5 = 35 + \frac{10}{3} \approx 38.33 \tag{1.2.10.4}$$

1.2.11 The larger of two supplementary angles exceeds the smaller by $18°$. Find the angles. (10 2019)

**Solution:** Let the larger angle be $x$ and the smaller angle be $y$. Since the two angles are supplementary, their sum is $180°$:

$$x + y = 180° \tag{1.2.11.1}$$

The larger angle exceeds the smaller by $18°$:

$$x - y = 18° \tag{1.2.11.2}$$

Expressing the system of linear equations in canonical matrix form $\mathbf{Mx} = \mathbf{b}$:

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 180^\circ \\ 18^\circ \end{pmatrix} \tag{1.2.11.3}$$

From (1.1.15.1), evaluating the inverse matrix where $\left|\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\right| = -1 - 1 = -2$:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 180^\circ \\ 18^\circ \end{pmatrix} \tag{1.2.11.4}$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 180^\circ \\ 18^\circ \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 198^\circ \\ 162^\circ \end{pmatrix} = \begin{pmatrix} 99^\circ \\ 81^\circ \end{pmatrix} \tag{1.2.11.5}$$

Therefore, the two supplementary angles are $99^\circ$ and $81^\circ$.

1.2.12 Sumit is 3 times as old as his son. Five years later, he shall be two and a half times as old as his son. How old is Sumit at present? (10 2019)

**Solution:** Let the present age of the son be $x$ years and Sumit's present age be $y$ years. From the given conditions:

$$y = 3x \implies 3x - y = 0 \tag{1.2.12.1}$$

Five years later, their ages will be $x + 5$ and $y + 5$ respectively:

$$y + 5 = \frac{5}{2}(x + 5) \implies 2y + 10 = 5x + 25 \implies 5x - 2y = -15 \tag{1.2.12.2}$$

Writing the system of equations in matrix form $\mathbf{Ax} = \mathbf{b}$:

$$\begin{pmatrix} 3 & -1 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ -15 \end{pmatrix} \tag{1.2.12.3}$$

From (1.1.15.1), since $\left|\mathbf{A}\right| = 3(-2) - (-1)(5) = -6 + 5 = -1$:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{-1} \begin{pmatrix} -2 & 1 \\ -5 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ -15 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 5 & -3 \end{pmatrix} \begin{pmatrix} 0 \\ -15 \end{pmatrix} = \begin{pmatrix} 15 \\ 45 \end{pmatrix} \tag{1.2.12.4}$$

Therefore, the present age of the son is 15 years and Sumit's present age is 45 years.

1.2.13 The probability of selecting a blue marble at random from a jar that contains only blue, black and green marbles is $\frac{1}{5}$. The probability of selecting a black marble at random from the same jar is $\frac{1}{4}$. If the jar contains 11 green marbles, find the total number of marbles in the jar. (10 2019)

**Solution:** Let the total number of marbles in the jar be $n$. Let $B$, $K$, and $G$ denote the events of selecting a blue, black, and green marble, respectively. Since these events partition the sample space, from (1.1.11.1):

$$\Pr(B) + \Pr(K) + \Pr(G) = 1 \tag{1.2.13.1}$$

Substituting $\Pr(B) = \frac{1}{5}$ and $\Pr(K) = \frac{1}{4}$:

$$\frac{1}{5} + \frac{1}{4} + \Pr(G) = 1 \implies \frac{9}{20} + \Pr(G) = 1 \implies \Pr(G) = \frac{11}{20} \tag{1.2.13.2}$$

By the classical definition of probability:

$$\Pr(G) = \frac{n(G)}{n} \implies \frac{11}{20} = \frac{11}{n} \implies n = 20 \tag{1.2.13.3}$$

Therefore, the total number of marbles in the jar is 20.

1.2.14 Find a relation between $x$ and $y$ if the points $A(x, y)$, $B(-4, 6)$ and $C(-2, 3)$ are collinear. (10 2019)

**Solution:** Let the position vectors of the given points be

$$\mathbf{A} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} -2 \\ 3 \end{pmatrix} \tag{1.2.14.1}$$

Forming the displacement vectors relative to point $\mathbf{B}$:

$$\mathbf{v}_1 = \mathbf{A} - \mathbf{B} = \begin{pmatrix} x - (-4) \\ y - 6 \end{pmatrix} = \begin{pmatrix} x + 4 \\ y - 6 \end{pmatrix} \tag{1.2.14.2}$$

$$\mathbf{v}_2 = \mathbf{C} - \mathbf{B} = \begin{pmatrix} -2 - (-4) \\ 3 - 6 \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix} \tag{1.2.14.3}$$

From (1.1.12.1), points $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are collinear if and only if the matrix formed by their displacement vectors has zero determinant:

$$\begin{vmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{vmatrix} = 0 \implies \begin{vmatrix} x + 4 & 2 \\ y - 6 & -3 \end{vmatrix} = 0 \tag{1.2.14.4}$$

Expanding the determinant:

$$-3(x + 4) - 2(y - 6) = 0 \implies -3x - 12 - 2y + 12 = 0 \implies 3x + 2y = 0 \tag{1.2.14.5}$$

The following code verifies the solution in Fig. 1.2.14.1:

codes/10/q14.py

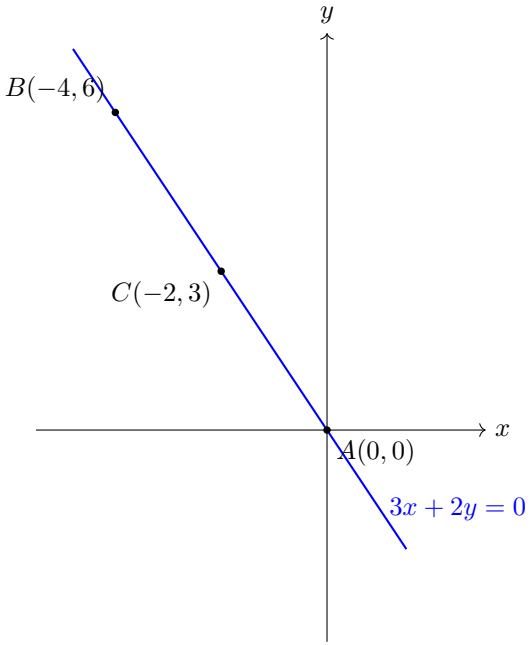



Fig. 1.2.14.1: Collinear points $A, B, C$ on the line $3x + 2y = 0$

1.2.15 Find the area of a triangle whose vertices are $(1, -1)$, $(-4, 6)$ and $(-3, -5)$. (10 2019)

**Solution:** Let the position vectors of the vertices be

$$\mathbf{A} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} -3 \\ -5 \end{pmatrix} \tag{1.2.15.1}$$

Forming the two displacement edge vectors from reference vertex $\mathbf{A}$:

$$\mathbf{u} = \mathbf{B} - \mathbf{A} = \begin{pmatrix} -4-1 \\ 6-(-1) \end{pmatrix} = \begin{pmatrix} -5 \\ 7 \end{pmatrix} \tag{1.2.15.2}$$

$$\mathbf{v} = \mathbf{C} - \mathbf{A} = \begin{pmatrix} -3-1 \\ -5-(-1) \end{pmatrix} = \begin{pmatrix} -4 \\ -4 \end{pmatrix} \tag{1.2.15.3}$$

From (1.1.13.1), the area of $\triangle ABC$ is:

$$\text{Area} = \frac{1}{2}\left|\left|\mathbf{u} \quad \mathbf{v}\right|\right| = \frac{1}{2}\left\|\begin{matrix} -5 & -4 \\ 7 & -4 \end{matrix}\right\| \tag{1.2.15.4}$$

$$= \frac{1}{2}\left|(-5)(-4) - (-4)(7)\right| = \frac{1}{2}\left|20 + 28\right| = \frac{1}{2}(48) = 24 \text{ sq. units} \tag{1.2.15.5}$$

The following code verifies the solution in Fig. 1.2.15.1:

codes/10/q15.py

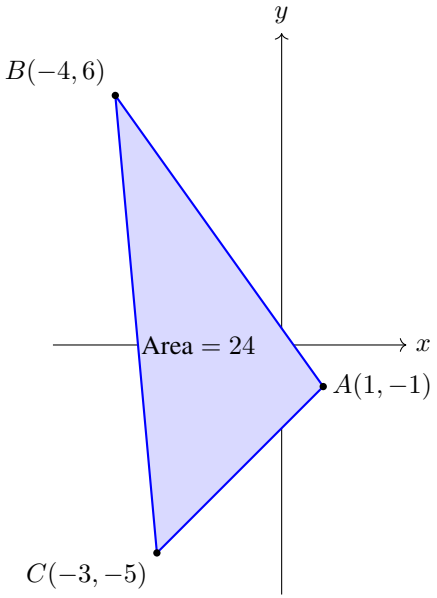



Fig. 1.2.15.1: Triangle $ABC$ with vertices $A(1,-1)$, $B(-4,6)$, and $C(-3,-5)$

1.2.16 Write the smallest number which is divisible by both 306 and 657. (10 2019)

**Solution:** The smallest positive integer divisible by both numbers is their least common multiple, $\text{lcm}\,(306,657)$. Applying Euclid's division lemma:

$$657 = 306 \times 2 + 45 \tag{1.2.16.1}$$
$$306 = 45 \times 6 + 36 \tag{1.2.16.2}$$
$$45 = 36 \times 1 + 9 \tag{1.2.16.3}$$
$$36 = 9 \times 4 + 0 \tag{1.2.16.4}$$

Since the remainder is 0, the greatest common divisor is:

$$\gcd\,(306,657) = 9 \tag{1.2.16.5}$$

From (1.1.7.1):

$$\text{lcm}\,(306,657) = \frac{306 \times 657}{\gcd\,(306,657)} \tag{1.2.16.6}$$
$$= \frac{306 \times 657}{9} = 34 \times 657 = 22338 \tag{1.2.16.7}$$

1.2.17 Point $A$ lies on the line segment $XY$ joining $X(6,-6)$ and $Y(-4,-1)$ in such a way that $\frac{XA}{XY} = \frac{2}{5}$. If point $A$ also lies on the line $3x + k(y+1) = 0$, find the value of $k$. (10 2019)

**Solution:** Point $A$ lies on line segment $XY$ with position vectors $\mathbf{X} = \begin{pmatrix} 6 \\ -6 \end{pmatrix}$ and

$\mathbf{Y} = \begin{pmatrix} -4 \\ -1 \end{pmatrix}$. Given $\frac{XA}{XY} = \frac{2}{5}$, since $XY = XA + AY$:

$$\frac{XA}{XA + AY} = \frac{2}{5} \implies 5XA = 2XA + 2AY \implies 3XA = 2AY \implies \frac{XA}{AY} = \frac{2}{3} \tag{1.2.17.1}$$

Thus, point $\mathbf{A}$ divides the line segment joining $\mathbf{X}$ and $\mathbf{Y}$ internally in the ratio $m : n = 2 : 3$. From (1.1.14.1):

$$\mathbf{A} = \frac{m\mathbf{Y} + n\mathbf{X}}{m + n} = \frac{2\begin{pmatrix} -4 \\ -1 \end{pmatrix} + 3\begin{pmatrix} 6 \\ -6 \end{pmatrix}}{2 + 3} \tag{1.2.17.2}$$

$$= \frac{1}{5}\left(\begin{pmatrix} -8 \\ -2 \end{pmatrix} + \begin{pmatrix} 18 \\ -18 \end{pmatrix}\right) = \frac{1}{5}\begin{pmatrix} 10 \\ -20 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \end{pmatrix} \tag{1.2.17.3}$$

Since point $\mathbf{A} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$ lies on the line $3x + k\,(y + 1) = 0$:

$$3\,(2) + k\,(-4 + 1) = 0 \implies 6 - 3k = 0 \implies 3k = 6 \implies k = 2 \tag{1.2.17.4}$$

The following code verifies the solution in Fig. 1.2.17.1:

codes/10/q17.py

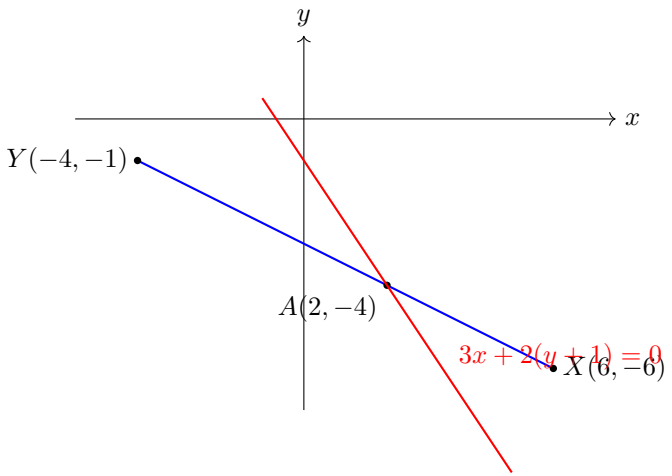



Fig. 1.2.17.1: Point $A(2, -4)$ dividing $XY$ in ratio $2 : 3$

1.2.18 Solve for $x$:

$$x^2 + 5x - \left(a^2 + a - 6\right) = 0 \tag{1.2.18.1}$$

(10 2019)

**Solution:** Comparing with the standard quadratic equation $Ax^2 + Bx + C = 0$:

$$A = 1, \quad B = 5, \quad C = -\left(a^2 + a - 6\right) \tag{1.2.18.2}$$

From (1.1.6.1), the discriminant $D$ is:

$$\begin{aligned} D = B^2 - 4AC &= 5^2 - 4\,(1)\,\left[-\left(a^2 + a - 6\right)\right] && (1.2.18.3)\\ &= 25 + 4\left(a^2 + a - 6\right) = 25 + 4a^2 + 4a - 24 && (1.2.18.4)\\ &= 4a^2 + 4a + 1 = (2a + 1)^2 && (1.2.18.5) \end{aligned}$$

Using the quadratic formula:

$$x = \frac{-B \pm \sqrt{D}}{2A} = \frac{-5 \pm \sqrt{(2a+1)^2}}{2\,(1)} = \frac{-5 \pm (2a+1)}{2} \tag{1.2.18.6}$$

Taking the positive sign:

$$x = \frac{-5 + (2a+1)}{2} = \frac{2a-4}{2} = a - 2 \tag{1.2.18.7}$$

Taking the negative sign:

$$x = \frac{-5 - (2a+1)}{2} = \frac{-2a-6}{2} = -\,(a+3) \tag{1.2.18.8}$$

Therefore, the roots are $x = a - 2$ or $x = -\,(a + 3)$.

1.2.19 Find $A$ and $B$ if $\sin(A + 2B) = \frac{\sqrt{3}}{2}$ and $\cos(A + 4B) = 0$, where $A$ and $B$ are acute angles. (10 2019)

**Solution:** From (1.1.2.1), since $\sin(60°) = \frac{\sqrt{3}}{2}$ and angles are acute:

$$A + 2B = 60° \tag{1.2.19.1}$$

Since $\cos(90°) = 0$:

$$A + 4B = 90° \tag{1.2.19.2}$$

Expressing the system of linear equations in matrix form $\mathbf{Mx} = \mathbf{b}$:

$$\begin{pmatrix} 1 & 2 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 60° \\ 90° \end{pmatrix} \tag{1.2.19.3}$$

From (1.1.15.1), evaluating the determinant and inverse of $\mathbf{M}$:

$$\left|\mathbf{M}\right| = \begin{vmatrix} 1 & 2 \\ 1 & 4 \end{vmatrix} = 1\,(4) - 2\,(1) = 2 \tag{1.2.19.4}$$

$$\mathbf{M}^{-1} = \frac{1}{2}\begin{pmatrix} 4 & -2 \\ -1 & 1 \end{pmatrix} \tag{1.2.19.5}$$

Solving for the angles:

$$\begin{pmatrix} A \\ B \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 4 & -2 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 60^\circ \\ 90^\circ \end{pmatrix} \tag{1.2.19.6}$$

$$= \frac{1}{2}\begin{pmatrix} 4\left(60^\circ\right) - 2\left(90^\circ\right) \\ -1\left(60^\circ\right) + 1\left(90^\circ\right) \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 240^\circ - 180^\circ \\ -60^\circ + 90^\circ \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 60^\circ \\ 30^\circ \end{pmatrix} = \begin{pmatrix} 30^\circ \\ 15^\circ \end{pmatrix} \tag{1.2.19.7}$$

Therefore, $A = 30^\circ$ and $B = 15^\circ$.

1.2.20 Prove that $2 + 5\sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is an irrational number. (10 2019)

**Solution:** Assume to the contrary that $2 + 5\sqrt{3}$ is a rational number. Then it can be written in the form $\frac{a}{b}$, where $a, b \in \mathbb{Z}$, $b \neq 0$, and $\gcd(a, b) = 1$:

$$2 + 5\sqrt{3} = \frac{a}{b} \tag{1.2.20.1}$$

Rearranging the terms:

$$5\sqrt{3} = \frac{a}{b} - 2 = \frac{a - 2b}{b} \tag{1.2.20.2}$$

$$\implies \sqrt{3} = \frac{a - 2b}{5b} \tag{1.2.20.3}$$

Since $a$ and $b$ are integers, $a - 2b$ and $5b$ are also integers with $5b \neq 0$. Thus, the right-hand side represents a rational number:

$$\frac{a - 2b}{5b} \in \mathbb{Q} \implies \sqrt{3} \in \mathbb{Q} \tag{1.2.20.4}$$

This contradicts the given fact that $\sqrt{3}$ is an irrational number. Hence, our assumption was false, and $2 + 5\sqrt{3}$ is an irrational number.

1.2.21 Using Euclid's Algorithm, find the HCF of 2048 and 960. (10 2019)

**Solution:** From (1.1.16.1), the reduction of an integer state vector under elementary unimodular shear matrices preserves the greatest common divisor:

$$\mathbf{v}_0 = \begin{pmatrix} 2048 \\ 960 \end{pmatrix} \tag{1.2.21.1}$$

Applying the first shear transformation $\mathbf{E}_1 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$:

$$\mathbf{v}_1 = \mathbf{E}_1\mathbf{v}_0 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2048 \\ 960 \end{pmatrix} = \begin{pmatrix} 2048 - 2\left(960\right) \\ 960 \end{pmatrix} = \begin{pmatrix} 128 \\ 960 \end{pmatrix} \tag{1.2.21.2}$$

Applying the second shear transformation $\mathbf{E}_2 = \begin{pmatrix} 1 & 0 \\ -7 & 1 \end{pmatrix}$:

$$\mathbf{v}_2 = \mathbf{E}_2\mathbf{v}_1 = \begin{pmatrix} 1 & 0 \\ -7 & 1 \end{pmatrix}\begin{pmatrix} 128 \\ 960 \end{pmatrix} = \begin{pmatrix} 128 \\ 960 - 7\left(128\right) \end{pmatrix} = \begin{pmatrix} 128 \\ 64 \end{pmatrix} \tag{1.2.21.3}$$

Applying the third shear transformation $\mathbf{E}_3 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$:

$$\mathbf{v}_3 = \mathbf{E}_3\mathbf{v}_2 = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 128 \\ 64 \end{pmatrix} = \begin{pmatrix} 128 - 2\,(64) \\ 64 \end{pmatrix} = \begin{pmatrix} 0 \\ 64 \end{pmatrix} \tag{1.2.21.4}$$

The non-zero invariant entry of the reduced state vector yields:

$$\gcd\,(2048, 960) = 64 \tag{1.2.21.5}$$

The following code verifies the solution:

1.2.22 Two right triangles $ABC$ and $DBC$ are drawn on the same hypotenuse $BC$ and on the same side of $BC$. If $AC$ and $BD$ intersect at $P$, prove that $AP \times PC = BP \times DP$. (10 2019)

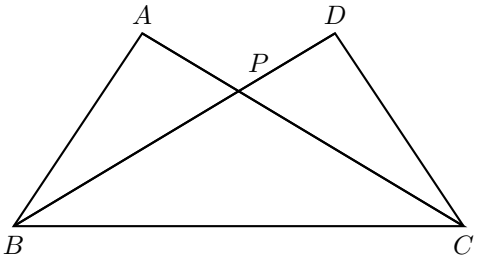



Fig. 1.2.22.1: Right triangles $ABC$ and $DBC$ on the common hypotenuse $BC$

**Solution:** In right triangles $ABC$ and $DBC$, both are right-angled with hypotenuse $BC$:

$$\angle BAC = 90^\circ, \quad \angle BDC = 90^\circ \tag{1.2.22.1}$$

Since the hypotenuse $BC$ subtends equal angles of $90^\circ$ at vertices $A$ and $D$ on the same side of $BC$, points $A, B, C, D$ are concyclic and lie on a circle with diameter $BC$. For the chord $AD$, angles in the same segment of the circle are equal:

$$\angle PBA = \angle PCD \tag{1.2.22.2}$$

In $\triangle PAB$ and $\triangle PDC$:

$$\angle APB = \angle DPC \quad \text{(vertically opposite angles)} \tag{1.2.22.3}$$
$$\angle PBA = \angle PCD \quad \text{(angles in the same segment)} \tag{1.2.22.4}$$

By the AA similarity criterion, $\triangle PAB \sim \triangle PDC$. From (1.1.17.1), the corresponding sides are proportional:

$$\frac{AP}{DP} = \frac{BP}{PC} \implies AP \times PC = BP \times DP \tag{1.2.22.5}$$

The following code verifies the geometric property:

1.2.23 Diagonals of a trapezium $PQRS$ intersect each other at the point $O$, $PQ \parallel RS$ and $PQ = 3RS$. Find the ratio of the areas of triangles $POQ$ and $ROS$. (10 2019)

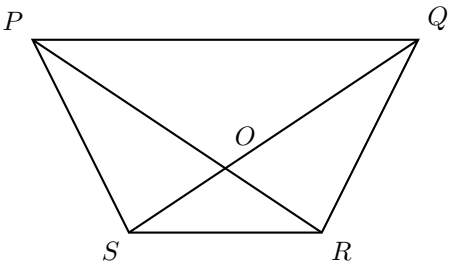



Fig. 1.2.23.1: Trapezium $PQRS$ with diagonals intersecting at $O$

**Solution:** In trapezium $PQRS$, given $PQ \parallel RS$. In $\triangle POQ$ and $\triangle ROS$:

$$\angle POQ = \angle ROS \quad \text{(vertically opposite angles)} \tag{1.2.23.1}$$
$$\angle OPQ = \angle ORS \quad \text{(alternate interior angles, since } PQ \parallel RS\text{)} \tag{1.2.23.2}$$
$$\angle OQP = \angle OSR \quad \text{(alternate interior angles, since } PQ \parallel RS\text{)} \tag{1.2.23.3}$$

Therefore, by the AAA similarity criterion, $\triangle POQ \sim \triangle ROS$. From (1.1.18.1):

$$\frac{\text{ar}\,(\triangle POQ)}{\text{ar}\,(\triangle ROS)} = \left(\frac{PQ}{RS}\right)^2 \tag{1.2.23.4}$$

Given that $PQ = 3RS$:

$$\frac{\text{ar}\,(\triangle POQ)}{\text{ar}\,(\triangle ROS)} = \left(\frac{3RS}{RS}\right)^2 = 3^2 = \frac{9}{1} \tag{1.2.23.5}$$

Thus, the ratio of the areas of $\triangle POQ$ and $\triangle ROS$ is $9 : 1$. The following code verifies the area ratio:

1.2.24 In Fig. 1.2.24.1, $PQ$ and $RS$ are two parallel tangents to a circle with centre $O$ and another tangent $AB$ with point of contact $C$ intersects $PQ$ at $A$ and $RS$ at $B$. Prove that $\angle AOB = 90°$. (10 2019)

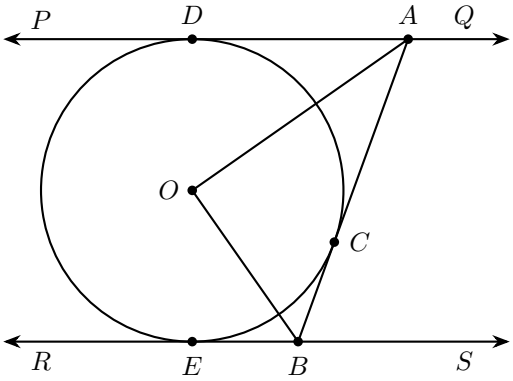



Fig. 1.2.24.1: Parallel tangents $PQ, RS$ and transversal tangent $AB$

**Solution:** Let the centre $O$ be the coordinate origin $\begin{pmatrix}0\\0\end{pmatrix}$. The parallel tangents $PQ$ and $RS$ correspond to the horizontal hyperplanes $y = r$ and $y = -r$. From (1.1.19.1), let the point of contact $C$ on the circle $\|\mathbf{x}\| = r$ be parameterized by:

$$\mathbf{r}_C = \begin{pmatrix} r\cos\theta \\ r\sin\theta \end{pmatrix}, \quad \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \tag{1.2.24.1}$$

The tangent line at $C$ with normal $\hat{n} = \begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix}$ has equation:

$$x\cos\theta + y\sin\theta = r \tag{1.2.24.2}$$

Intersecting with $y = r$ (line $PQ$) and $y = -r$ (line $RS$):

$$\mathbf{r}_A = \begin{pmatrix} r\frac{1-\sin\theta}{\cos\theta} \\ r \end{pmatrix}, \quad \mathbf{r}_B = \begin{pmatrix} r\frac{1+\sin\theta}{\cos\theta} \\ -r \end{pmatrix} \tag{1.2.24.3}$$

From (1.1.19.2), evaluating the scalar product $\mathbf{r}_A^\top \mathbf{r}_B$:

$$\mathbf{r}_A^\top \mathbf{r}_B = r^2 \left(\frac{1-\sin\theta}{\cos\theta}\right)\left(\frac{1+\sin\theta}{\cos\theta}\right) + r\,(-r) \tag{1.2.24.4}$$

$$= r^2\left(\frac{1-\sin^2\theta}{\cos^2\theta}\right) - r^2 = r^2\,(1) - r^2 = 0 \tag{1.2.24.5}$$

Since $\mathbf{r}_A^\top \mathbf{r}_B = 0$, $\mathbf{OA} \perp \mathbf{OB}$, which proves that $\angle AOB = 90°$. The following code verifies the orthogonality:

1.2.25 In Fig. 1.2.25.1, a square $OABC$ is inscribed in a quadrant $OPBQ$. If $OA = 15$ cm, find the area of the shaded region. (Use $\pi = 3.14$.) (10 2019)

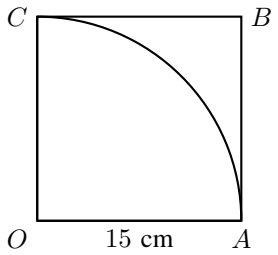



Fig. 1.2.25.1: Square $OABC$ inscribed in circular quadrant $OPBQ$

**Solution:** From (1.1.20.2), the side length of square $OABC$ is $s = OA = 15$ cm. Applying Pythagoras theorem (1.1.4.1) in right-angled $\triangle OAB$, diagonal $OB$ forms the radius $r$ of quadrant $OPBQ$:

$$r^2 = OB^2 = OA^2 + AB^2 = 15^2 + 15^2 = 450 \implies r = 15\sqrt{2} \text{ cm} \tag{1.2.25.1}$$

From (1.1.20.1), the area of quadrant $OPBQ$ is:

$$\text{Area of quadrant} = \frac{1}{4}\pi r^2 = \frac{1}{4}(3.14)(450) = \frac{1413}{4} = 353.25 \text{ cm}^2 \quad (1.2.25.2)$$

The area of square $OABC$ is:

$$\text{Area of square} = s^2 = 15^2 = 225 \text{ cm}^2 \quad (1.2.25.3)$$

Therefore, the area of the shaded region is:

$$\text{Area of shaded region} = \text{Area of quadrant} - \text{Area of square} \quad (1.2.25.4)$$

$$= 353.25 - 225 = 128.25 \text{ cm}^2 \quad (1.2.25.5)$$

The following code verifies the solution:

1.2.26 In Fig. 1.2.26.1, $ABCD$ is a square with side $2\sqrt{2}$ cm and inscribed in a circle. Find the area of the shaded region. (Use $\pi = 3.14$.) (10 2019)

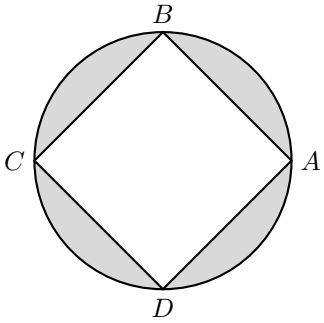



Fig. 1.2.26.1: Square $ABCD$ inscribed in circumscribed circle

**Solution:** Let the side of the inscribed square $ABCD$ be $s = 2\sqrt{2}$ cm. Since each interior angle of the square is $90°$, diagonal $AC$ is the diameter of the circle. From (1.1.4.1):

$$AC^2 = AB^2 + BC^2 = \left(2\sqrt{2}\right)^2 + \left(2\sqrt{2}\right)^2 = 8 + 8 = 16 \implies AC = 4 \text{ cm} \quad (1.2.26.1)$$

Thus, the radius of the circle is:

$$r = \frac{AC}{2} = \frac{4}{2} = 2 \text{ cm} \quad (1.2.26.2)$$

From (1.1.20.3), the area of the circle is:

$$\text{Area of circle} = \pi r^2 = 3.14 \times 2^2 = 12.56 \text{ cm}^2 \quad (1.2.26.3)$$

The area of the inscribed square is:

$$\text{Area of square} = s^2 = \left(2\sqrt{2}\right)^2 = 8 \text{ cm}^2 \quad (1.2.26.4)$$

The area of the shaded region is:

$$\text{Area of shaded region} = \text{Area of circle} - \text{Area of square} \tag{1.2.26.5}$$

$$= 12.56 - 8 = 4.56 \text{ cm}^2 \tag{1.2.26.6}$$

The following code verifies the solution:

1.2.27 A solid is in the form of a cylinder with hemispherical ends. The total height of the solid is 20 cm and the diameter of the cylinder is 7 cm. Find the total volume of the solid. (Use $\pi = \frac{22}{7}$.) (10 2019)

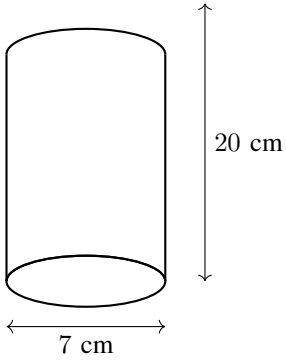



Fig. 1.2.27.1: Cylinder with hemispherical ends

**Solution:** The diameter is $d = 7$ cm, giving radius $r = \frac{7}{2}$ cm. Total height of the solid is $H = 20$ cm. The height of the cylindrical body $h$ is:

$$h = H - 2r = 20 - 2\left(\frac{7}{2}\right) = 20 - 7 = 13 \text{ cm} \tag{1.2.27.1}$$

From (1.1.21.1), the volume of the cylindrical portion is:

$$V_{\text{cyl}} = \pi r^2 h = \frac{22}{7} \times \left(\frac{7}{2}\right)^2 \times 13 = \frac{22}{7} \times \frac{49}{4} \times 13 = \frac{1001}{2} = 500.5 \text{ cm}^3 \tag{1.2.27.2}$$

From (1.1.21.2), the two hemispherical ends combine to form a sphere:

$$V_{\text{ends}} = 2 \times \left(\frac{2}{3}\pi r^3\right) = \frac{4}{3}\pi r^3 = \frac{4}{3} \times \frac{22}{7} \times \left(\frac{7}{2}\right)^3 = \frac{539}{3} \approx 179.67 \text{ cm}^3 \tag{1.2.27.3}$$

The total volume of the solid is:

$$V = V_{\text{cyl}} + V_{\text{ends}} = \frac{1001}{2} + \frac{539}{3} = \frac{3003 + 1078}{6} = \frac{4081}{6} \approx 680.17 \text{ cm}^3 \tag{1.2.27.4}$$

The following code verifies the computed volume:

```
codes/10/q27.py
```

1.2.28 The marks obtained by 100 students in an examination are given below:

| Marks | 30–35 | 35–40 | 40–45 | 45–50 | 50–55 | 55–60 | 60–65 |
|---|---|---|---|---|---|---|---|
| Number of Students | 14 | 16 | 28 | 23 | 18 | 8 | 3 |

Find the mean marks of the students. (10 2019)

**Solution:** Constructing the computation table with class marks $x_i = \frac{l_i+u_i}{2}$:

| Class Interval | Class Mark ($x_i$) | Frequency ($f_i$) | $f_i x_i$ |
|---|---|---|---|
| 30–35 | 32.5 | 14 | 455.0 |
| 35–40 | 37.5 | 16 | 600.0 |
| 40–45 | 42.5 | 28 | 1190.0 |
| 45–50 | 47.5 | 23 | 1092.5 |
| 50–55 | 52.5 | 18 | 945.0 |
| 55–60 | 57.5 | 8 | 460.0 |
| 60–65 | 62.5 | 3 | 187.5 |
| **Total** | | $\sum f_i = 110$ | $\sum f_i x_i = 4930.0$ |

From (1.1.22.1), the arithmetic mean is:

$$\bar{x} = \frac{\sum_{i=1}^{k} f_i x_i}{\sum_{i=1}^{k} f_i} = \frac{4930}{110} = \frac{493}{11} \approx 44.82 \tag{1.2.28.1}$$

The following code verifies the solution in Fig. **??**:

```
codes/10/q28.py
```

1.2.29 For what value of $k$ is the polynomial $f(x) = 3x^4 - 9x^3 + x^2 + 15x + k$ completely divisible by $3x^2 - 5$? (10 2019)

**Solution:** From (1.1.23.1), work in the quotient ring $\mathbb{R}[x]/\langle 3x^2 - 5\rangle$, where the divisor induces the reduction relation:

$$3x^2 - 5 \equiv 0 \implies x^2 \equiv \frac{5}{3} \tag{1.2.29.1}$$

Higher powers of the state variable $x$ reduce algebraically:

$$x^3 = x \cdot x^2 \equiv \frac{5}{3}x \tag{1.2.29.2}$$

$$x^4 = \left(x^2\right)^2 \equiv \left(\frac{5}{3}\right)^2 = \frac{25}{9} \tag{1.2.29.3}$$

Projecting $f(x)$ modulo $3x^2 - 5$:

$$f(x) \equiv 3\left(\frac{25}{9}\right) - 9\left(\frac{5}{3}x\right) + \left(\frac{5}{3}\right) + 15x + k \tag{1.2.29.4}$$

$$\equiv \frac{25}{3} - 15x + \frac{5}{3} + 15x + k \tag{1.2.29.5}$$

$$\equiv \frac{30}{3} + k \equiv 10 + k \pmod{3x^2 - 5} \tag{1.2.29.6}$$

For $f(x)$ to be completely divisible, the remainder residue must be zero:

$$10 + k = 0 \implies k = -10 \tag{1.2.29.7}$$

The following code verifies the polynomial division:

codes/10/q29.py

1.2.30 Find the zeroes of the quadratic polynomial $7y^2 - \frac{11}{3}y - \frac{2}{3}$ and verify the relationship between the zeroes and the coefficients. (10 2019)

**Solution:** To find the zeroes, set $p(y) = 7y^2 - \frac{11}{3}y - \frac{2}{3} = 0$. Multiplying by 3:

$$21y^2 - 11y - 2 = 0 \tag{1.2.30.1}$$

Factoring by splitting the middle term:

$$21y^2 - 14y + 3y - 2 = 0 \tag{1.2.30.2}$$

$$7y\,(3y - 2) + 1\,(3y - 2) = 0 \tag{1.2.30.3}$$

$$(3y - 2)\,(7y + 1) = 0 \tag{1.2.30.4}$$

Equating each factor to zero:

$$3y - 2 = 0 \implies y = \frac{2}{3}, \qquad 7y + 1 = 0 \implies y = -\frac{1}{7} \tag{1.2.30.5}$$

Thus, the zeroes are $\alpha = \frac{2}{3}$ and $\beta = -\frac{1}{7}$.
Comparing $p(y)$ with the standard quadratic polynomial $ay^2 + by + c$:

$$a = 7, \quad b = -\frac{11}{3}, \quad c = -\frac{2}{3} \tag{1.2.30.6}$$

Verifying Vieta's relations (1.1.24.1) and (1.1.24.2):

$$\alpha + \beta = \frac{2}{3} + \left(-\frac{1}{7}\right) = \frac{14 - 3}{21} = \frac{11}{21} = -\frac{-11/3}{7} = -\frac{b}{a} \tag{1.2.30.7}$$

$$\alpha\beta = \left(\frac{2}{3}\right)\left(-\frac{1}{7}\right) = -\frac{2}{21} = \frac{-2/3}{7} = \frac{c}{a} \tag{1.2.30.8}$$

Hence, the relationship between zeroes and coefficients is verified.

1.2.31 Prove that the ratio of the areas of two similar triangles is equal to the ratio of the squares on their corresponding sides. (10 2019)

**Solution:** Let $\triangle ABC \sim \triangle PQR$. Draw altitudes $AM \perp BC$ and $PN \perp QR$. The ratio of their areas is:

$$\frac{\text{ar}\,(\triangle ABC)}{\text{ar}\,(\triangle PQR)} = \frac{\frac{1}{2}BC \times AM}{\frac{1}{2}QR \times PN} = \left(\frac{BC}{QR}\right)\left(\frac{AM}{PN}\right) \tag{1.2.31.1}$$

In $\triangle ABM$ and $\triangle PQN$, $\angle B = \angle Q$ (as $\triangle ABC \sim \triangle PQR$) and $\angle AMB = \angle PNQ = 90^\circ$. By AA similarity:

$$\triangle ABM \sim \triangle PQN \implies \frac{AM}{PN} = \frac{AB}{PQ} \tag{1.2.31.2}$$

Since $\triangle ABC \sim \triangle PQR$, the corresponding sides are proportional:

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP} \tag{1.2.31.3}$$

Substituting $\frac{AM}{PN} = \frac{BC}{QR}$ yields (1.1.18.1):

$$\frac{\text{ar}\,(\triangle ABC)}{\text{ar}\,(\triangle PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{BC}{QR}\right)^2 = \left(\frac{CA}{FD}\right)^2 \tag{1.2.31.4}$$

1.2.32 Two poles of equal heights are standing opposite to each other on either side of the road which is 80 m wide. From a point $P$ between them on the road, the angle of elevation of the top of a pole is $60^\circ$ and the angle of depression from the top of the other pole to point $P$ is $30^\circ$. Find the heights of the poles and the distance of the point $P$ from the poles. (10 2019)

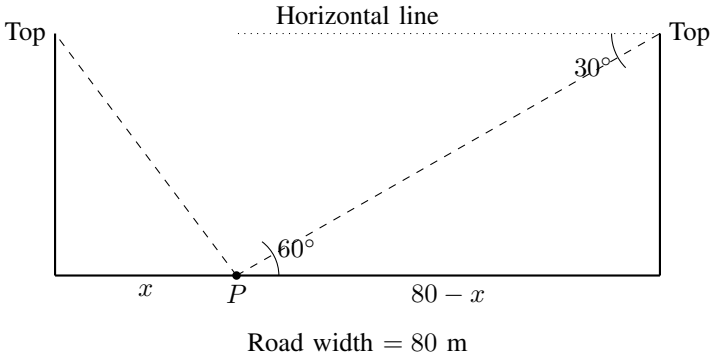



Fig. 1.2.32.1: Poles and point of observation on road

**Solution:** Let the height of both poles be $h$ and the distance of $P$ from the first pole be $x$. From the angles of elevation and depression:

$$\tan(60^\circ) = \frac{h}{x} \implies h = \sqrt{3}x \tag{1.2.32.1}$$

$$\tan(30^\circ) = \frac{h}{80-x} \implies h = \frac{80-x}{\sqrt{3}} \tag{1.2.32.2}$$

Equating (1.2.32.1) and (1.2.32.2):

$$\sqrt{3}x = \frac{80-x}{\sqrt{3}} \implies 3x = 80 - x \implies 4x = 80 \implies x = 20 \text{ m} \tag{1.2.32.3}$$

The height of each pole is:

$$h = 20\sqrt{3} \text{ m} \tag{1.2.32.4}$$

The distance from the second pole is $80 - 20 = 60$ m.

1.2.33 The total cost of a certain length of a piece of cloth is Rs. 200. If the piece was 5 m longer and each metre of cloth costs Rs. 2 less, the cost of the piece would have

remained unchanged. How long is the piece and what is its original rate per metre? (10 2019)

**Solution:** Let the original length of cloth be $x$ metres. The original rate per metre is $\frac{200}{x}$. With the new conditions:

$$(x+5)\left(\frac{200}{x}-2\right)=200 \tag{1.2.33.1}$$

Expanding:

$$200-2x+\frac{1000}{x}-10=200 \tag{1.2.33.2}$$

$$-2x-10+\frac{1000}{x}=0 \tag{1.2.33.3}$$

$$x^2+5x-500=0 \tag{1.2.33.4}$$

Factoring:

$$(x+25)(x-20)=0 \implies x=20 \text{ m} \quad (\text{since } x>0) \tag{1.2.33.5}$$

The original rate is:

$$\text{Rate}=\frac{200}{20}=10 \text{ Rs./m} \tag{1.2.33.6}$$

1.2.34 Which term of the Arithmetic Progression $-7$, $-12$, $-17$, $-22$, ... will be $-82$? Is $-100$ any term of the A.P.? Give reason for your answer. (10 2019)

**Solution:** Here, $a=-7$ and $d=-12-(-7)=-5$. From (**??**):

$$a_n=a+(n-1)\,d \tag{1.2.34.1}$$

Setting $a_n=-82$:

$$-82=-7+(n-1)(-5) \implies -75=-5(n-1) \implies n=16 \tag{1.2.34.2}$$

Thus, the $16^{\text{th}}$ term is $-82$. Now testing $-100$:

$$-100=-7+(m-1)(-5) \implies -93=-5(m-1) \implies m=1+\frac{93}{5}=19.6 \tag{1.2.34.3}$$

Since $m \notin \mathbb{Z}k$, $-100$ is not a term of the A.P. The following code verifies the solution:

1.2.35 Change the following data into a "less than type" distribution and draw its ogive:

| Class Interval | 30–40 | 40–50 | 50–60 | 60–70 | 70–80 | 80–90 | 90–100 |
|---|---|---|---|---|---|---|---|
| Frequency | 7 | 5 | 8 | 10 | 6 | 6 | 8 |

(10 2019)

**Solution:** The cumulative frequency table for the "less than type" distribution using the upper class limits:

| Marks (Upper Limit) | Frequency ($f$) | Cumulative Frequency ($cf$) |
|---|---|---|
| Less than 40 | 7 | 7 |
| Less than 50 | 5 | 12 |
| Less than 60 | 8 | 20 |
| Less than 70 | 10 | 30 |
| Less than 80 | 6 | 36 |
| Less than 90 | 6 | 42 |
| Less than 100 | 8 | 50 |
| **Total** | $N = 50$ | |

Plotting points $(30, 0)$, $(40, 7)$, $(50, 12)$, $(60, 20)$, $(70, 30)$, $(80, 36)$, $(90, 42)$, $(100, 50)$ on the Cartesian plane yields the ogive curve. Here, total observations $N = 50$:

$$\frac{N}{2} = \frac{50}{2} = 25 \tag{1.2.35.1}$$

Locating 25 on the $y$-axis and projecting onto the ogive curve, the corresponding value on the $x$-axis gives the median:

$$\text{Median} = 65 \tag{1.2.35.2}$$

1.2.36 Draw two concentric circles of radii 2 cm and 5 cm. Take a point $P$ on the outer circle and construct a pair of tangents $PA$ and $PB$ to the smaller circle. Measure $PA$. (10 2019)

**Solution:** Let the common centre be $\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. Point $\mathbf{P}$ lies on the outer circle such that $\|\mathbf{P}\| = 5$ cm. Since $PA$ is tangent to the inner circle of radius $r = 2$ cm at point $A$, radius $\mathbf{OA}$ is orthogonal to tangent $\mathbf{PA}$ ($\mathbf{OA} \perp \mathbf{PA}$). Applying Pythagoras theorem (1.1.4.1) in right-angled $\triangle OAP$:

$$PA^2 = OP^2 - OA^2 = 5^2 - 2^2 = 25 - 4 = 21 \implies PA = \sqrt{21} \approx 4.58 \text{ cm} \tag{1.2.36.1}$$

The following code generates the geometric construction:

codes/10/q36.py

1.2.37 A solid iron pole consists of a cylinder of height 220 cm and base diameter 24 cm, which is surmounted by another cylinder of height 60 cm and radius 8 cm. Find the mass of the pole, given that 1 cm$^3$ of iron has approximately 8 gm mass. (Use $\pi = 3.14$.) (10 2019)

**Solution:** The iron pole consists of two coaxial right circular cylinders stacked vertically.

Lower (Larger) Cylinder:

$$h_1 = 220 \text{ cm}, \quad r_1 = \frac{24}{2} = 12 \text{ cm} \tag{1.2.37.1}$$

From (1.1.21.1), the volume of the lower cylinder is:

$$V_1 = \pi r_1^2 h_1 = 3.14 \times 12^2 \times 220 = 3.14 \times 144 \times 220 = 99475.2 \text{ cm}^3 \tag{1.2.37.2}$$

Upper (Smaller) Cylinder:

$$h_2 = 60 \text{ cm}, \quad r_2 = 8 \text{ cm} \tag{1.2.37.3}$$

The volume of the upper cylinder is:

$$V_2 = \pi r_2^2 h_2 = 3.14 \times 8^2 \times 60 = 3.1464 \times 60 = 12057.6 \text{ cm}^3 \tag{1.2.37.4}$$

Total Volume:

$$V = V_1 + V_2 = 99475.2 + 12057.6 = 111532.8 \text{ cm}^3 \tag{1.2.37.5}$$

Total Mass: From (1.1.21.3), with density $\rho = 8$ gm/cm$^3$:

$$\text{Mass} = V \times \rho = 111532.8 \times 8 = 892262.4 \text{ gm} = 892.26 \text{ kg} \tag{1.2.37.6}$$

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